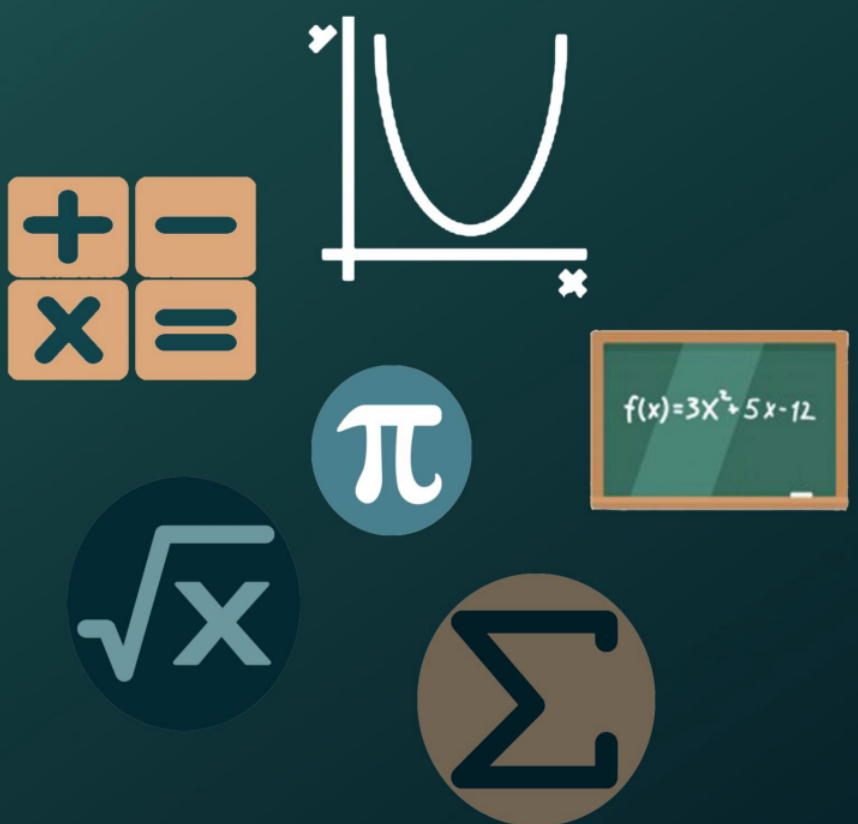
MATHEMATICS

Quadratic Equations & Expressions



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Polynomials				19
2	Quadratic equation and Nature of Roots	1-5	1, 15, 19 - 23	1	23,24
3	Formation of Quadratic Equations	6-8	11-12, 24		20
4	Symmetric Function of Roots	8	25		
5	Roots Under Special Conditions	9-11	5-7		
6	Identities	12	13, 26		1,25
7	Condition of Common Roots	13-15	2, 16, 27	2	26,27
8	Quadratic Graphs	16-19	11, 17, 28,	3	
9	Ineuqalities	20- 24	3		2,28
10	Maximum and Minimum Values of Quadratic and Rational Functions	25-27	8 - 10, 14, 18	4 - 6	29,30
11	Location of Roots		12, 29	7 - 9	4-12,17,18, 21,31-34
12	Transformation of Quadratic Equations	28-29			
13	Theory of Equations	30-32	30	10-12	13-16,35
14	Quadratic Equation in Two Variables		4		
15	Modulus Function	33-34			36
16	Modulus Inequalities	35		13	
17	Logarithmic Inequalities	36	13, 31, 33		3,22,37
18	Exponential Inequalities	37-39			
19	ACC				38

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Quadratic Equation & Nature of Roots

Q1. If one root of equation $x^2 - px + 5 = 0$ is 2, then value of p is ?

(A) –2 (B) $\frac{7}{2}$

(C) $\frac{9}{2}$ (D) None

Q2. If the roots of the equation $x^2 - bx + c = 0$ and $x^2 - cx + b = 0$ differ by the same quantity, then b + c is equals to ?$(b \neq c)$

(A) –4 (B) 1

(C) 0 (D) 4

Q3. If $2 + i\sqrt{3}$ is a root of equation $x^2 + px + q = 0$ then the value of (p, q) is : $(p, q \in R)$

(A) (–7, 4) (B) (–4, 7)

(C) (4, –7) (D) (7, –4)

Q4. The roots of the equation $x^2 - 2\sqrt{2}x + 1 = 0$ are :

(A) Imaginary and different

(B) Real and different

(C) Real and equal

(C) Rational and different

Q5. The number of roots of the quadratic equation $8\sec^2\theta - 6\sec\theta + 1 = 0$ are:

(A) Infinite (B) 1

(C) 2 (D) 0

Formation of Quadratic Equation and Symmetric Functions

Q6. If the roots of equation $ax^2 + bx + c = 0$ are λ and 2λ, then:

(A) $b^2 = 9ac$ (B) $2b^2 = 9ac$

(C) $b^2 = -4ac$ (D) $a^2 = c^2$

Q7. If the roots of the equation $x^2 + px + q = 0$ are tan 22° and tan 23° then :-

(A) p + q = 1 (B) p + q = –1

(C) p – q = 1 (D) p – q = –1

Q8. If α, β are the roots of $x^2 - 4x + 1 = 0$ then $\alpha^3 + \beta^3$ is :

(A) 76 (B) –76

(C) 52 (D) –52

Roots under special conditions

Q9. The value of p for which the quadratic equation $x^2 - px + p + 3 = 0$ has reciprocal roots is :

(A) 1 (B) –1

(C) –2 (D) None

Q10. The quadratic equation whose roots are $\tan^2\theta, \cot^2\theta$ can be (where $\theta \neq \frac{n\pi}{2}, n \in I$).

(A) $x^2 - 2x + 1 = 0$

(B) $x^2 - 3x + 1 = 0$

(C) $x^2 - 5x + 1 = 0$

(D) All of them

Q11. If $p > 0, q > 0$ then roots of $x^2 - px - q = 0$ are :-

(A) Imaginary

(B) Real and of opposite sign

(C) Real and positive

(D) Real and negative

Identities

Q12. The value of 'a' for which
$(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + a(a + 1) = 0$
is satisfied by more than two distinct values of x, is
(A) 1 (B) 0
(C) –1 (D) none of these

Condition for common roots

Q13. If the equation $x^2 + 4x + 6 = 0$ and $ax^2 + bx + c = 0$ a, b, c ∈ R have a common root, then a : b : c is :
(A) 1 : 4 : 6 (B) 6 : 4 : 1
(C) 1 : 2 : 3 (D) None

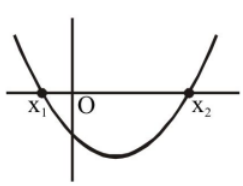
Q14. The value of b for which the equations $x^2 + bx + 1 = 0$, $x^2 + x + b = 0$ have one real root in common is :-
(A) –2 (B) 1 (C) $\sqrt{2}$ (D) $\sqrt{3}$

Q15. If a, b, c ∈ R and equations $ax^2 + bx + c = 0$ and $2x^2 + 3x + 4 = 0$ have a common root, then $\frac{a+c}{b}$ is equals to :-
(A) $\frac{1}{2}$ (B) 2 (C) $\frac{5}{4}$ (D) $\frac{4}{3}$

Quadratic Graphs

Q16. If $y = f(x) = ax^2 + 2bx + c = 0$ has imaginary roots and $4a + 4b + c < 0$ then :-
(A) c > 0 (B) c < 0
(C) c = 0 (D) Data insufficiant

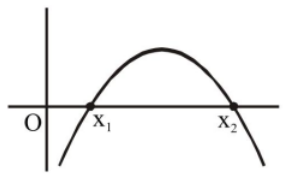
Q17. The following figure shows the graph of $y = ax^2 + bx + c$, then :



(A) a < 0 (B) c < 0
(C) $b^2 < 4ac$ (D) none of these

Q18. If c > 0 and the equation $3ax^2 + 4bx + c = 0$ has no real root, then :-
(A) 2a + c > b (B) a + 2c > b
(C) 3a + c > 4b (D) a + 3c < b

Q19. The following figure shows the graph of $y = ax^2 + bx + c$ then
(A) a > 0
(B) $b^2 < 4ac$
(C) c > 0
(D) b > 0



Inequalities

Q20. Solve for x $\frac{x^2+6x-7}{x^2+1} \le 2$:-
(A) x ∈ R (B) (–1, 3)
(C) [–1, 3] (D) None of these

Q21. Solve for 'x' if $\frac{1-2x-3x^2}{3x-x^2-5} > 0$:-
(A) $(-\infty, 1)$
(B) $\left(\frac{1}{3}, \infty\right)$
(C) $(-\infty, 2)$
(D) $(-\infty, -1) \cup \left(\frac{1}{3}, \infty\right)$

Q22. Solution set of x for which
$\frac{(x-1)(x-2)^2(x+4)}{(x+2)(x-3)} \ge 0$ is-
(A) $(-\infty, -4] \cup (-2, 1] \cup \{2\} \cup (3, \infty)$
(B) $(-4, 2] \cup [1, 2] \cup (3, \infty)$
(C) $(-\infty, -4] \cup \{2\} \cup (3, \infty)$
(D) None of these

Q23. The number of integral solution of
$\frac{x+2}{x^2+1} > \frac{1}{2}$ is -
(A) 0 (B) 1
(C) 2 (D) 3

Q24. If $3x^2 + ax + 3 > 0$, $\forall\ x \in R$ then :-
(A) $a \in (-6, 0)$ (B) $a \in (-6, 6)$
(C) $a \in [-6, 6]$ (D) $a \in (0, 6)$

Maximum and Minimum Values of Quadratic and Rational Functions

Q25. If x be real, then the minimum value of $x^2 - 8x + 17$ is :
(A) –1 (B) 0
(C) 2 (D) 1

Q26. The greatest value of the expression $\frac{1}{4x^2+2x+1}$ is :
(A) $\frac{4}{3}$ (B) $\frac{5}{2}$
(C) $\frac{13}{14}$ (D) none of these

Q27. If a, b are non zero distinct roots of $x^2 + ax + b = 0$ then the least value of $x^2 + ax + b$ is :
(A) 1 (B) $-\frac{2}{3}$
(C) $-\frac{9}{4}$ (D) None

Transformation and Theory of Equations

Q28. If $\alpha\ \beta$ are root's of equation $ax^2 + bx + c = 0$, the equation whose roots are $2 + \alpha$, $2 + \beta$ is:
(A) $ax^2 + (4a - b)x + 4a - 2b + c = 0$
(B) $ax^2 + (4a - b)x + 4a + 2b + c = 0$
(C) $ax^2 + (b - 4a)x + 4a - 2b + c = 0$
(D) None

Q29. If α, β are the roots of $ax^2 + bx + c = 0$, then the roots of equation $ax^2 - bx(x-1) + c(x-1)^2 = 0$ are :-
(A) $\frac{\alpha}{\alpha-1}, \frac{\beta}{\beta-1}$ (B) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$
(C) $\frac{\alpha+1}{\alpha}, \frac{\beta+1}{\beta}$ (D) α, β

Q30. If α, β, γ are the roots of the equation $x^3 + 4x + 1 = 0$, then $(\alpha + \beta)^{-1} + (\beta + \gamma)^{-1} + (\gamma + \alpha)^{-1}$ equals :-
(A) 2 (B) 3 (C) 4 (D) 5

Q31. If α, β, γ are the roots of $x^3 - x + 1 = 0$ then the equation whose roots are $\alpha + \beta$, $\beta + \gamma$ and $\gamma + \alpha$ is:
(A) $x^3 + x + 1 = 0$ (B) $x^3 - x - 1 = 0$
(C) $x^3 + x - 1 = 0$ (D) None

Q32. Let α and β be the two disinct roots of the equation $x^3 + 3x^2 - 1 = 0$. The equation which has $(\alpha\beta)$ as its root is equal to
(A) $x^3 - 3x - 1 = 0$
(B) $x^3 - 3x^2 + 1 = 0$
(C) $x^3 + x^2 - 3x + 1 = 0$
(D) $x^3 + x^2 + 3x - 1 = 0$

Modulus Function

Q33. The number of real roots of equation $x^2 + 5|x| + 6 = 0$ are :
(A) 0 (B) 1
(C) 2 (D) None

Q34. The sum of the real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is
(A) 1 (B) –1 (C) 2 (D) 4

Inequalities involving Modulus Functions

Q35. Number of integral values of x satisfying the inequliaty $\frac{x^2+6x-7}{|x+4|} < 0$ is :-
(A) 5 (B) 6 (C) 7 (D) 8

Logarithmic Inequalities

Q36. Solve, $\log_{0.5}\frac{3-x}{x+2}<0$:-

(A) $x\in(-2,0)$ (B) $x\in\left(-2,\frac{1}{2}\right)$

(C) $x\in\left(0,\frac{1}{2}\right)$ (D) $x\in(-2,\infty)$

Exponental Inequalities

Q37. The number of real roots of $3^{2x^2-7x+7}=9$ is :

(A) 0 (B) 1 (C) 2 (D) 4

Q38. The number of real solution of equation $\left(\frac{3}{2}\right)^x=-x^2+5x-10$:-

(A) 1 (B) 2

(C) 4 (D) 0

Q39. If $x=\sqrt{6+\sqrt{6+\sqrt{6+}}}$..... then :

(A) $x=-2$

(B) $x=3$

(C) $x=-2, 3$

(D) None

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. For which value of c, given quadratic equation has rational roots :-

$2x^2-7x+c=0$

(A) $c=0$ (B) $c=3$

(C) $c=6$ (D) $c=9$

Q2. If equation $x^2-5x+6=0$ and equation $x^2-6x+8=0$ has only one root in common then :-

(A) common root is 2

(B) sum of other root of equation is 2

(C) sum of other root of equation is 7

(D) common root is 3

Q3. If S is the set of all real number x such that $\frac{x^2-8x+7}{x^3-3x^2+2x}$ is negative then subsets of S is/are :-

(A) (1, 7) (B) (2, 7)

(C) $(-\infty, 0)$ (D) $(0, \infty)$

Q4. If $x^2+2xy+2x+my-3$ have two linear factor then m is :-

(A) –2 (B) 2

(C) 6 (D) –6

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 5 to 7)

$f(x)=x^2+2(k+1)x+9k-5$

Q5. The values of k such that f(x) has real zeros is/are -

(A) $k\le 1$ (B) $k\ge 6$

(C) $k\ge 1$ (D) $k\le 6$

Q6. The values of k such that f(x) = 0 has roots of opposite sign :-

(A) $k \le 0$ (B) $k \ge 0$

(C) $0 \le k \le 1/2$ (D) $k \ge 1/2$

Q7. If f(x) = (x – k)(x – 10) + 1, then values of 'k' for which f(x) = 0 has integral roots :-

(A) k = 6 (B) k = 8

(C) k =10 (D) k = 12

Paragraph # 2 (Ques. 8 to 10)

Let $f(x) = 4x^2 - 4ax + a^2 - 2a + 2$ be a quadratic polynomial in x, a be any real number.

On the basis of above information, answer the following questions:

Q8. If x- coordinate of vertex of parabola y = f(x) is less than 0 and f(x) has minimum value 3 for $x \in [0, 2]$, then value of a is :

(A) $1+\sqrt{2}$ (B) $1-\sqrt{2}$

(C) $1-\sqrt{3}$ (D) $1+\sqrt{3}$

Q9. If y = f(x) takes minimum value 3 on [0, 2] and x- coordinate of vertex is greater than 2, then value of a is -

(A) $5-\sqrt{10}$ (B) $10-\sqrt{5}$

(C) $5+\sqrt{10}$ (D) $10+\sqrt{5}$

Q10. If at least one root of f(x) = 0 lies in [0, 2], then the value of a belongs to :

(A) $\left[5-\sqrt{7}, 5+\sqrt{7}\right]$

(B) $\left[1, 5+\sqrt{7}\right]$

(C) $(-5, +5) \cup (5+, \infty)$

(D) $(-5, \infty)$

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. Four options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q11. Consider a quadratic equation $f(x) = x^2 + kx + 4$, now

Column-I	**Column-II**
(i) Graph of y = f(x) intersects x-axis	(P) $k \in (-\infty, -4) \cup (4, \infty)$
(ii) Graph of y = f(x) touches x-axis	(Q) $k \in (-4, 4)$
(iii) Graph of y = f(x) Neither cut nor touches	(R) $k \in \{-4, 4\}$
(iv) $f(x) > 0 \ \ x \in R$	

(A) (i)→(Q), (ii)→(R), (iii)→(P), (iv)→(Q)

(B) (i)→(P), (ii)→(R), (iii)→(Q), (iv)→(Q)

(C) (i)→(R), (ii)→(Q), (iii)→(P), (iv)→(R)

(D) (i)→(P), (ii)→(Q), (iii)→(R), (iv)→(R)

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q12. Consider the equation $x^2 + 2(a-1)x + a + 5 = 0$, where 'a' is a parameter. Match of the real values of 'a' so that the given equation has :

Column I	**Column II**
(A) imaginary roots	(P) $\left(-\infty, -\frac{8}{7}\right)$
(B) one root smaller than 3 and other root greater than 3	(Q) $(-1, 4)$
(C) exactly one root in the interval (1, 3) & 1 and 3 are not the root of the equation	(R) $\left(-\frac{4}{3}, -\frac{8}{7}\right)$
(D) one root smaller than 1and other root greater than 3	(S) $\left(-\infty, -\frac{4}{3}\right)$

Q13.

Column-I	**Column-II**
(A) The set of real value of x for which $\log_{0.2} \frac{x+2}{x} \le 1$ is	(P) $[2, 4]$
(B) The solution set of the inequality $\log_{10}(x^2-16) \le \log_{10}(4x-11)$	(Q) $\left(0, \frac{\sqrt{5}-1}{2}\right]$
(C) If $\log_{\cos x} \sin x \ge 2$ and $0 \le x \le 3\pi$ then sin x lie in the interval	(R) $(4, 5)$
(D) The set of real values of x satisfying $\log_{1/2}(x^2-6x+12) \ge -2$ is	(S) $\left(-\infty, \frac{-5}{2}\right) \cup (0, +\infty)$

TRIPLE MATCH TYPE

*This section contains **ONE table (each having 3 columns and 4 rows)** Each option in Column I will match with **ONLY ONE OPTION** in Column II and Column III. For example one possible answer can be (A, R, H).*

Q14.

Column IColumn II	**Column III**	
y = f(x)	Roots of f(x) = 0 are real	$f(x) > 0 \forall x \in R$
(A) $x^2 - 6x + k$	(P) $k \ge 9$	(E) $k > 16$
(B) $x^2 - 8x + k$	(Q) $k \le 16$	(F) $k \in (-16, 16)$
(C) $x^2 - kx + 64$	(R) $k \ge 16$	(G) $k > 9$
(D) $x^2 - 2kx + 81$	(S) $k \le 9$	(H) $k \in (-9, 9)$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q15. if 'a' is least possible positive integes and $a,b,c \in I$ such that equation $ax^2 + bx + c = 0$ also $a + b + c = 0$ and product of roots is 1/4 then value of $2(a + c) + b$ is :-

Q16. The value of 'a' for which $x^3 + ax - 1 = 0$ & $x^4 + ax^2 + 1 = 0$ have a common root is_____.

Q17. Let $f(x) = ax^2 + bx + c$, $a, b, c \in R$ and given that vertex of the quadratic equation is at $(1, -4)$ then find $|a + b + c|$:-

Q18. Minimum value of $\log_2(x^2 - 4x + 8)$ is $\forall x \in R$.

SUBJECTIVE TYPE

Q19. If the roots of $x^2 - bx + c = 0$ are the consecutive integers find $b^2 - 4c$.

Q20. α, β are the roots of the equation $K(x^2 - x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.

Q21. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$.

Q22. If α, β are the roots of $ax^2 + bx + c = 0$, $(a \neq 0)$ and $\alpha + \delta$, $\beta + \delta$ are the roots of $Ax^2 + Bx + C = 0$, $(A \neq 0)$ for some constant δ, then prove that, $\dfrac{b^2 - 4ac}{a^2} = \dfrac{B^2 - 4AC}{A^2}$.

Q23. If the roots of $x^2 - ax + b = 0$ are real & differ by a quantity which is less than $c (c > 0)$, prove that b lies between $(1/4)(a^2 - c^2)$ & $(1/4)a^2$.

Q24. Form the quadratic equations with rational coefficients. whose one root is $\tan 15^\circ$.

(Note: $\tan 15^\circ = 2 - \sqrt{3}$)

Q25. (a) If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$ then which of the following expressions in α, β will denote the symmetric functions of roots. Give proper reasoning.

(i) $f(\alpha, \beta) = \alpha^2 - \beta$

(ii) $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$

(iii) $f(\alpha, \beta) = \ln \dfrac{\alpha}{\beta}$

(iv) $f(\alpha, \beta) = \cos(\alpha - \beta)$

(b) If α, β are the roots of the equation $x^2 - px + q = 0$, then find the quadratic equation the roots of which are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ & $\alpha^3\beta^2 + \alpha^2\beta^3$.

Q26. If $\dfrac{(a+x)^2}{(a-b)(a-c)} + \dfrac{(b+x)^2}{(b-c)(b-a)} + \dfrac{(c+x)^2}{(c-a)(c-b)} = 1$ then prove that it is an identity.

Q27. If $Q_1(x) = x^2 + (k - 29)x - k$ and $Q_2(x) = 2x^2 + (2k - 43)x + k$ both are factors of a cubic polynomial $P(x)$, then the largest value of k is -

Q28. The quadratic equation $ax^2 + bx + c = 0$ has no real root, then prove that $c(a + b + c) > 0$.

Q29. Solve the cubic $4x^3 + 16x^2 - 9x - 36 = 0$, the sum of its two roots being equal to zero.

Q30. $\log_2 \log_4 \log_5 x > 0$ then find value of x.

Q31. Solve for x :

$\log_{0.5}(x^2 - 5x + 6) \geq -1$

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The roots of the quadratic equation

$(a+b-2c)x^2-(2a-b-c)x+(a-2b+c)=0$

are -

(A) a + b + c & a – b + c

(B) 1/2 & a – 2b + c

(C) a – 2b + c & 1/(a + b – 2c)

(D) None of these

Q2. If $x^2 + ax + 10 = 0$ and $x^2 + bx - 10 = 0$ have a common root, then $a^2 - b^2$ is equal to -

(A) 10 (B) 20

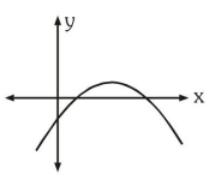
(C) 30 (D) 40

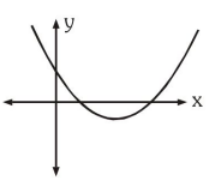
Q3. Graph of the function $f(x) = Ax^2 - BX + C$, where

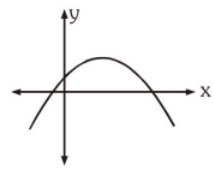
$A = (\sec\theta - \cos\theta)(\operatorname{cosec}\theta - \sin\theta)(\tan\theta + \cot\theta)$,

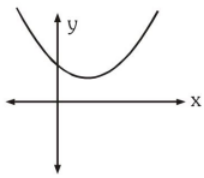
$B = (\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 - (\tan^2\theta + \cot^2\theta)$ &

C = 12, is represented by

(A) 

(B) 

(C) 

(D) 

Q4. If $x \in R$ then maximum and minimum value of $\frac{x^2-6x+4}{x^2+2x+4}$ are :-

(A) $5, -\frac{1}{3}$ (B) $1, -\frac{1}{3}$

(C) $7, \frac{1}{7}$ (D) 4, 1

Q5. If a, b, $c \in R$ and the equation $ax^2 + bx + c = 0$, $a \neq 0$, has real roots α and β satisfying $\alpha < -1$ and $\beta > 1$ then $1 + \frac{c}{a} + \left|\frac{b}{a}\right|$ is :

(A) Positive (B) Negative

(C) Zero (D) None

Q6. If $x^2 + 2ax + a < 0 \ \forall \ x \in [1, 2]$ then :-

(A) $a \in (-\infty, -\frac{4}{5})$ (B) $a \in (-\frac{4}{5}, -\frac{1}{3})$

(C) $a \in (-\infty, -\frac{1}{3})$ (D) None of these

Q7. Let a, b, $c \in R$ and α, β are the real roots of the equation $ax^2 + bx + c = 0$ and if $a < 0$, $b > 0$, $c > 0$ and $\alpha < \beta$ then :-

(A) $\alpha < 0 < \beta < |\beta|$

(B) $\alpha < 0 < \beta < |\alpha|$

(C) $\beta < 0 < \alpha < |\beta|$

(D) $\alpha < 0 < |\alpha| < \beta$

Q8. Both the roots of the equation $x^2 - 2kx + k^2 - 1 = 0$ lie between -2 & 4, then k belongs to the interval :-

(A) $(-1, 1)$ (B) $(1, 3)$

(C) $(-1, 3)$ (D) $(-3, -1)$

Q9. If the roots of equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3 then :

(A) $a < 2$ (B) $2 \le a \le 3$

(C) $3 < a \le 4$ (D) $a > 4$

Q10. If two roots of the equation $x^3 - px^2 + qx - r = 0$ are equal in magnitude but opposite in sign, then :-

(A) $pr = q$ (B) $qr = p$

(C) $pq = r$ (D) None of these

Q11. If $\alpha, \beta, \gamma, \delta$ are roots of equation $x^4 - Kx^3 + Kx^2 + Lx + M = 0$ where $K, L, M \in R$ then minimum value of $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is :-

(A) 0 (B) -1

(C) 1 (D) 2

Q12. If α, β, γ are the roots of $x^3 + x + 1 = 0$ then the value of $\alpha^3 + \beta^3 + \gamma^3$ is :

(A) 0 (B) 3

(C) -3 (D) None

Q13. If $\frac{1-|x|}{2-|x|} \ge 0$ then :-

(A) $x \in (-\infty, \infty) - [-2, 2]$

(B) $x \in (-\infty, \infty) - [-1, 1]$

(C) $x \in [-1,1] \cup (-\infty, -2) \cup (2, \infty)$

(D) $x \in (-\infty, \infty) - [-2, 1]$

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\alpha_1 < \alpha_2 < \alpha_3 < \alpha_4 < \alpha_5 < \alpha_6$, then the equation $(x - \alpha_1)(x - \alpha_3)(x - \alpha_5) + 3(x - \alpha_2)(x - \alpha_4)(x - \alpha_6) = 0$ has :-

(A) three real roots

(B) no real root in $(-\infty, \alpha_1)$

(C) one real root in (α_1, α_2)

(D) no real root in (α_5, α_6)

Q2. If $a > 2$ is a constant & there are 28 positive integers satisfying $(x - a)(x - 2a)(x - a^2) < 0$, then

(A) 'a' is composite

(B) 'a' is odd

(C) $a > 8$

(D) $a \in (3, 11)$

Q3. The equation $x^{3/4(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$ has

(A) at least one real root

(B) exactly three real solution

(C) exactly one irrational solution

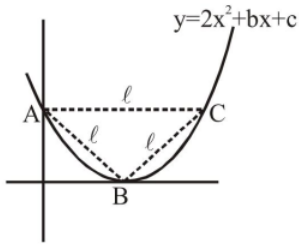
(D) exactly one integral solution

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 4 to 6)

In the given figure, a parabola $y = 2x^2 + bx + c$ touches x-axis at vertex B and cuts y-axis at A. A line parallel to x-axis is drawn cutting the parabola at A & C such that ΔABC is equilateral with side length equal to ℓ.



On the basis of above information, answer the following questions :

Q4. The area of ΔABC (in square units) is

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{6\sqrt{3}}{4}$ (C) $\frac{\sqrt{3}}{8}$ (D) $\frac{\sqrt{3}}{4}$

Q5. Number of integral values of 'k' for which $\frac{k}{2}$ lies between b & c is :-

(A) 8 (B) 9 (C) 10 (D) 11

Q6. Cubic equation whose roots are 2ℓ, b and c is

(A) $x^3-\frac{3}{2}x^2+12x+18=0$

(B) $x^3-\frac{3}{2}x^2+12x-18=0$

(C) $x^3-\frac{3}{2}x^2-12x+18=0$

(D) $x^3-\frac{3}{2}x^2-12x-18=0$

Paragraph # 2 (Ques. 7 to 9)

Consider the quadratic equation
$(1+k)x^2-2(1+2k)x+(3+k)=0$,
where $k \in R-\{-1\}$.

Q7. The number of integral values of k such that the given quadratic equation has imaginary roots are:

(A) 0 (B) 1
(C) 2 (D) 3

Q8. The set of values of k such that the given quadratic has both the roots positive is :

(A) $k \in R$

(B) $k \in (-\infty,-3) \cup [\sqrt{2/3},\infty)$

(C) $k \in (-\infty,-3) \cup (-1,\infty)$

(D) $k \in (-\infty,-\sqrt{2/3}] \cup [\sqrt{2/3},\infty)$

Q9. The number of real values of k such that the given quadratic equation has roots in the ratio 1 : 2 is

(A) 0 (B) 1 (C) 2 (D) ∞

Paragraph # 3 (Ques. 10 to 12)

Consider a rational function $f(x)=\frac{x^2-3x-4}{x^2-3x+4}$ and a quadratic function $g(x) = x^2-(b+1)x+b-1$, where b is a parameter.

Q10. The sum of integers in the range of f(x) is -

(A) – 5 (B) – 6
(C) – 9 (D) – 10

Q11. If both roots of the equation g(x) = 0 are greater than – 1, then b lies in the interval.

(A) $(-\infty,-2)$ (B) $\left(-\infty,\frac{-1}{4}\right)$

(C) $(-2,\infty)$ (D) $\left(\frac{-1}{2},\infty\right)$

Q12. The largest natural number b satisfying $g(x) > -2 \ \forall \ x \in R$, is -

(A) 1 (B) 2 (C) 3 (D) 4

Paragraph # 4 (Ques. 13 to 16)

If α, β, γ be the roots of the equation $ax^3+bx^2+cx+d=0$. To obtain the equation whose roots are f(α), f(β), f(γ), where f is a function, we put y = f(α) and simplify it to obtain α = g(y) (some function of y). Now, α is a root of the equation $ax^3+bx^2+cx+d=0$, then we obtain the desired equation which is $a\{g(y)\}^3+b\{g(y)\}^2+c\{g(y)\}+d=0$.

For example, if α, β, γ are the roots of $ax^3+bx^2+cx+d=0$. To find equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$

we put $y=\frac{1}{\alpha} \Rightarrow \alpha=\frac{1}{y}$

As α is a root of $ax^3+bx^2+cx+d=0$

we get $\frac{a}{y^3}+\frac{b}{y^2}+\frac{c}{y}+d=0 \Rightarrow dy^3+cy^2+by+a=0$

This is the desired equation.

Q13. If α, β are the roots of the equation $ax^2+bx+c=0$, then the roots of the equation $a(2x+1)^2+b(2x+1)(x-1)+c(x-1)^2=0$ are :

(A) $\frac{2\alpha+1}{\alpha-1}, \frac{2\beta+1}{\beta-1}$ (B) $\frac{2\alpha-1}{\alpha+1}, \frac{2\beta-1}{\beta+1}$

(C) $\frac{\alpha+1}{\alpha-2}, \frac{\beta+1}{\beta-2}$ (D) $\frac{2\alpha+3}{\alpha-1}, \frac{2\beta+3}{\beta-1}$

Q14. If α, β are the roots of the equation $2x^2+4x-5=0$, the equation whose roots are the reciprocals of $2\alpha-3$ and $2\beta-3$ is :

(A) $x^2+10x-11=0$

(B) $11x^2+10x+1=0$

(C) $x^2+10x+11=0$

(D) $11x^2-10x+1=0$

Q15. If α, β are the roots of the equation $px^2-qx+r=0$, then the equation whose roots are $\alpha^2+\frac{r}{p}$ and $\beta^2+\frac{r}{p}$ is :

(A) $p^3x^2+pq^2x+r=0$

(B) $px^2-qx+r=0$

(C) $p^3x^2-pq^2x+q^2r=0$

(D) $px^2+qx-r=0$

Q16. If α, β, γ are the roots of the equation $x^3-x-1=0$, then the value of $\prod\left(\frac{1+\alpha}{1-\alpha}\right)$ is equal to -

(A) -7 (B) -5 (C) -3 (D) -1

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17.

Column-I	**Column-II**
(A) The maximum value of $\frac{x^2-6x+4}{x^2+2x+4}$ (x is real) is	(P) 0
(B) The correct valeu of a for which the equation $(a^2+4a+3)x^2+(a^2-a-2)x+a(a+1)=0$ has more than two root is	(Q) 1
(C) The number of real values of x satisfying $5^x+5^{-x}=\log_{10}25$ is	(R) -1
	(S) $-1/3$
	(T) 5

Q18. Consider the quadratic trinomial $f(x) = 2x^2 - 10px + 7p - 1$, where p is a parameter. Find the range of p in the following conditions given in column-I.

Column-I	Column-II
(A) If both roots of $f(x) = 0$ are confined in $(-1, 1)$ then	(P) $\left(\frac{2}{5}, \infty\right)$
(B) Exactly one root of $f(x) = 0$ lies in $(-1, 1)$	(Q) ϕ
(C) Both roots of $f(x) = 0$ are greater than 1	(R) $\left(-\frac{1}{17}, \frac{1}{3}\right)$
(D) One root of $f(x) = 0$ is greater than 1 and other root of $f(x) = 0$ is less than -1	(S) $\left(-\infty, -\frac{1}{17}\right] \cup \left[\frac{1}{3}, \infty\right)$
	(T) $\left(-\infty, -\frac{1}{17}\right) \cup \left(\frac{1}{3}, \infty\right)$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. The value of expression $x^4 - 8x^3 + 18x^2 - 8x + 2$ when $x = 2 + \sqrt{3}$ is _____.

Q20. If $f(x) = x^2 - 4ax + 5a^2 - 6a$, then largest distance between the zeroes of $f(x)$.

Q21. The value of "a" for which all roots of quadratic equation, $f(x) = (a-2)x^2 + 2ax + a + 3 = 0$ lies in $(-2, 1)$ belongs to $\left(-\infty, -\frac{1}{4}\right) \cup (m, n]$ then value of $n - m$ is ?

Q22. If sum of all solutions of the equation $(x^{\log_{10} 3})^2 - (3^{\log_{10} x}) - 2 = 0$ is $a^{\log_b c}$ where $a,b,c \in N$ & a, b are prime numbers then $a + b$ equals.

SUBJECTIVE TYPE

Q23. Show that if p, q, r, s are real numbers and $pr = 2(q + s)$, then at least one of the equations $x^2 + px + q = 0$ and $x^2 + rx + s = 0$ has real roots.

Q24. If roots of the equation $(x - \alpha)(x - 4 + \beta) + (x - 2 + \alpha)(x + 2 - \beta) = 0$ are p and q then find the absolute value of the sum of the roots of the equations $2(x - p)(x - q) - (x - \alpha)(x - 4 + \beta) = 0$ and $2(x - p)(x - q) - (x - 2 + \alpha)(x + 2 - \beta) = 0$.

Q25. For what values of p, the equation $(p + 2)(p - 1)x^2 + (p - 1)(2p + 1)x + p^2 - 1 = 0$ has more than two roots.

Q26. Suppose a cubic polynomial $f(x) = x^3 + px^2 + qx + 72$ is divisible by both $x^2 + ax + b$ and $x^2 + bx + a$ (where a, b, p, q are constants and $a \neq b$). Find the sum of the squares of the roots of the cubic polynomial.

Q27. Let $P(x) = x^2 + bx + c$, where b and c are integer. If $P(x)$ is a factor of both $x^4 + 6x^2 + 25$ and $3x^4 + 4x^2 + 28x + 5$, find the value of $P(1)$.

Q28. Let $f(x)=\frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$.

Solve the following inequality :

(i) $f(x) > 0$ (ii) $f(x) \geq 0$

(iii) $f(x) < 0$ (iv) $f(x) \leq 0$

Q29. We call 'p' a good number if the inequality $\frac{2x^2+2x+3}{x^2+x+1} \leq p$ is satisfied for any real x.

Find the smallest integral good number.

Q30. Let $P(x) = 4x^2 + 6x + 4$

and $Q(y) = 4y^2 - 12y + 25$.

Find the unique pair of real numbers (x, y) that satisfy $P(x) \cdot Q(y) = 28$.

Q31. For what value of 'p' the vertex of the parabola if $x^2 + 2px + 13$ lies at a distance of 5 unit from the origin.

Q32. For the quadratic equation

$x^2-(m-3)x+m=0$,

find the value of m for which

(i) one root is smaller than 2 and the other is greater than 2

(ii) both roots are grater than 2

(iii) both roots lie in (1, 2)

(iv) exactly one root lie in (1, 2)

Q33. For what real values of 'a' do the roots of $x^2 - 2x - a^2 + 1 = 0$ lie between the roots $x^2 - 2(a + 1)x + a(a - 1) = 0$.

Q34. Find the complete set of real values of 'a' for which both roots of the quadratic equation $(a^2 - 6a + 5)x^2 - \sqrt{a^2+2a}\,x + (6a - a^2 - 8) = 0$ lie on either side of the origin.

Q35. $\alpha, \beta, \gamma, \delta$ are the roots of the equation $\tan\left(\frac{\pi}{4}+x\right) = 3\tan 3x$ no two of which have equal tangents, find the value of $\tan\alpha + \tan\beta + \tan\gamma + \tan\delta$.

Q36. Solve the following equation $|2x - 3| - |x^2 - 4x + 3| = |x^2 - 2x|$.

Q37. Solve the inequation :

$\log_{2x+3} x^2 < \log_{2x+3}(2x+3)$.

Q38. Let a, b, c be positive integers and consider all the quadratic equations of the form $ax^2 - bx + c = 0$ which have two distinct real roots in the open interval $0 < x < 1$. Find the least positive integer a for which such a quadratic equation exists.

Answer Key

Ex - 1									
1	2	3	4	5	6	7	8	9	10
C	A	B	B	D	B	D	C	C	D
11	12	13	14	15	16	17	18	19	20
B	C	A	A	B	B	B	C	D	A
21	22	23	24	25	26	27	28	29	30
D	A	D	B	D	A	C	C	B	C
31	32	33	34	35	36	37	38	39	
B	A	A	D	B	B	C	D	B	

Ex - 1A

1. A,B,C **2.** A,C **3.** B,C **4.** A,C **5.** A,B

6. A,C **7.** B,D **8.** B **9.** B **10.** B

11. B

12. A $\rightarrow$ Q, B $\rightarrow$ P, C $\rightarrow$ R, D $\rightarrow$ S

13. A $\rightarrow$ P,Q,R,S; B $\rightarrow$ R, R; C$\rightarrow$ Q, Q; D $\rightarrow$ P

14. (A, S, G) ; (B, Q, E); (C, R, F) ; (D, P, H)

15. 5 **16.** –2 **17.** 4 **18.** 2 **19.** 1

20. 254 **21.** 20 **24.** $x^2 - 4x + 1 = 0$

25. (a) (ii) and (iv) ; (b) $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

27. k = 30 **29.** $-\frac{3}{2}, \frac{3}{2}, -4$. **30.** x > 625 **31.** $x \in [1,2) \cup (3,4]$

Ex - 2									
1	2	3	4	5	6	7	8	9	10
D	D	B	A	B	A	D	C	A	C
11	12	13							
B	C	C							

Ex - 2A

1. A,B,C **2.** A,D **3.** A,B,C,D **4.** A **5.** B

6. C **7.** B **8.** B **9.** C **10.** B

11. D **12.** B **13.** C **14.** B **15.** C

16. D

17. (A) $\rightarrow$ (T); (B) $\rightarrow$ (R) ; (C) $\rightarrow$ (P)

18. (A) $\rightarrow$ (R); (B) $\rightarrow$ (S) ; (C) $\rightarrow$ (Q) ; (D) $\rightarrow$ (Q)

19. 1 **20.** 6 **21.** 1 **22.** 5 **24.** 4

25. 1 **26.** 146 **27.** 4

28. (i) $x \in (-\infty, -6) \cup (1, 3) \cup (7, \infty)$,

(ii) $x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$

(iii) $x \in (-6, -2) \cup (-2, 0) \cup (0, 1) \cup (3, 7)$,

(iv) $x \in [-6, 0) \cup (0, 1] \cup [3, 7) \cup \{2\}$

29. 4 **30.** $\left(-\frac{3}{4}, \frac{3}{2}\right)$ **31.** $p = \pm 3, \pm 4$

32. (i) $m \in (10, \infty)$, (ii) $m \in [9, 10)$, (iii) $m \in \phi$, (iv) $m \in (10, \infty)$

33. $\frac{-1}{4} < a < 1$ **34.** $\therefore a \in (-\infty, -2] \cup [0,1) \cup (2,4) \cup (5, \infty)$

35. 0 **36.** $x \in [0, 1] \cup [2, 3]$

37. $x \in \left(-\frac{3}{2}, -1\right) \cup (-1, 0) \cup (0, 3)$ **38.** 5

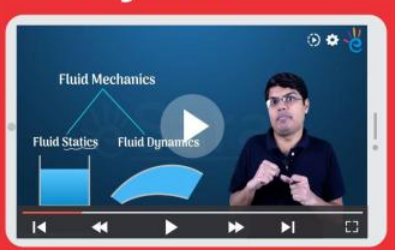
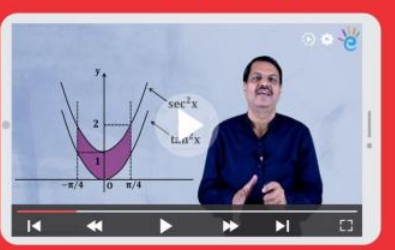
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